

MACHINE LEARNING MODEL FOR PREDICTIVE FILE ANALYSIS



A Capstone Project Presented to the Faculty of the
College of Computer Studies St. Michael's College
Iligan City

By

**AWA, REYNAN B.
CUA, JUSTINE A.**

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Abstract

This research focuses on developing a machine learning-based system for detecting malicious PDF files specifically those containing hidden or obfuscated JavaScript code. The study aims to create a predictive model that classifies PDF files as either benign or malicious. Machine learning methods such as Random Forest and Logistic Regression were evaluated for their ability to accurately classify PDFs, with Logistic Regression showing slightly better performance in detecting malicious files. A dynamic analysis framework using Docker was also developed to simulate the behavior of PDF files and identify threats that only appear during execution. The findings revealed that both machine learning models performed effectively, with Logistic Regression achieving an 86.67% accuracy in detecting malicious PDFs and Random Forest achieving 80%. The dynamic analysis framework proved useful for detecting links and threats. The study concludes that machine learning models combined with dynamic analysis techniques can enhance real-time detection of malicious PDFs.

149 words

Keywords: Pdf, obfuscated, machine learning



DEDICATION

This journey has been an absolute rollercoaster, full of treasured moments and memories that will endure forever. I want to dedicate this study to the people who have supported me the most during this journey.

First of all, I would like to thank God for allowing me to be strong in the spirit, wise in the mind, and above all for being there during this trying but fulfilling process. This work is in memory of all who have supported me from start to the end, first my parents who loved me unconditionally, sacrificed so much for me thereby becoming my strength.

On this particular occasion of thanksgiving, I want to thank you for your mentorship and support that have remained consistent even during the best, and worst times of my life and career, through the good, bad and the ugly. Your kindness has been my recovery and healing and I have used your favors to navigate this path.

-Reynan



DEDICATION

To my family, friends, and mentors, whose unwavering support and encouragement have been my source of strength throughout this journey.

To my parents, for their unconditional love and belief in my abilities; to my friends, for providing me with moments of relief and joy during challenging times; and to my mentors and professors, for their guidance and wisdom, which have been invaluable in shaping my academic path.

This work is also dedicated to the countless hours spent overcoming challenges in training the machine learning model, particularly in optimizing algorithms and handling complex datasets. The limited availability of high-quality resources and computational power tested my perseverance and adaptability, making this accomplishment all the more meaningful.

Lastly, I dedicate this thesis to the spirit of learning and discovery, which drives innovation and progress.

-Justine



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We thank Ma'am Fe, Sir Veldin and Ma'am MJ for their suggestions to make our system more efficient through their valuable comments. Without these insights and ideas, there is no good way to modify and shape the system. To the researchers family; we thank you for your whole hearted and untiring support-financially and emotionally-and for strengthening our patience and courage to strive harder for the fulfillment of this study,

First and foremost, to our Almighty God, who is our sustainer in our endeavors, provider of our needs, and our keeper of the bonds that unite us until the realization of the manuscript.

-Reynan Awa & Justine Cua



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LIST OF ACRONYMS AND ABBREVIATIONS

GNOME	GNU Network Object Model Environment
AI	Artificial Intelligence
ML	Machine Learning





Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

Malware intended to damage computer systems is a serious concern today due to its ability to alter, remove, or steal data for negative purposes. These programs rely on known signatures to find threats. However, if new malware doesn't match any known signatures this method often fails to spot them [1].

One way to spread malware is to attach it in PDF files. PDF stands for Portable Document Format. It was developed in 1993 by Adobe Inc. It includes executable code, images, text, and other attachments [2]. Researchers at Forcepoint have found that hackers are using these dangerous PDFs to target people who purchase tickets online [3]. Another exploit found by EdgeSpot where almost all anti-virus engines failed to detect was an obfuscation technique with steganography. The sample includes two layers of obfuscation. The code pretends to be harmless pictures labeled as icons but is actually decoded and run using JavaScript in the PDF [4].

Moreover, these cybercriminals hide harmful Word documents inside innocent-looking PDF files. These files can look like different types depending on the program used to open them. When these PDFs are opened, they use VBS macros to quietly install malware on computers by taking advantage of



weaknesses in programs like Microsoft Office [5]. These vulnerabilities allow attackers to run harmful code when a PDF is opened. Hackers often focus on outdated or unpatched PDF readers and the software's key parts. This can lead to unauthorized access to systems or stolen data. Most common malware on PDFs are exploits. The exploit is used by the malware to gain access to a system through known security vulnerabilities. When a pdf is opened it drops a file where it scans the network and attempts to spread [6].

There are two ways of analyzing malware in PDF files. These are static analysis and dynamic analysis. Static analysis is struggling to find malware with obfuscated code. However, dynamic analysis which tracks behavior in real time, static analysis looks at the code without running it and focuses on features. These programs have trouble with packaged malware that encrypts its code in order to avoid detection. Static analysis fails to detect obfuscated code that is only revealed during execution [7].

The researchers want to make PDF files safer by improving how we detect hidden malware especially with obfuscated code. They'll use dynamic analysis, which watches how PDF files behave in real time. These effective methods include using Docker and machine learning. The system focuses on PDF files where users can upload in a web-based system to identify malware and analyze files.



Statement of the Problem

There are three problems that have been identified by the researchers:

1. Users trust PDF files as safe because they appear as static documents, which gives hackers an advantage in hiding malicious code built in JavaScript that can be exploited.
2. Static analysis cannot evaluate code that is being executed in real-time, which makes it difficult to identify malware that only shows itself during runtime
3. Users are unable to use safe PDF files when antivirus programs flag safe files as dangerous threats.

Objective of the Study

Creating a machine learning model for predictive file analysis is the aim of this research. This study specifically seeks to;

1. Create a machine learning model to detect if a pdf file is malicious or benign with an extraction feature to identify obfuscated JavaScript code.
2. Develop a dynamic analysis framework using Docker to simulate PDF file behavior and structural analysis.
3. Evaluate machine learning methods such as Random Forest and Logistic Regression to identify malicious PDF files.



Scope and Limitation of the Study

The study will focus only on applying supervised learning to identify if a PDF file is malicious or not. These methods include Random Forest and Logistic Regression to train the model using datasets from Contagio that is a well-known collection of malware samples accessible via the Contagio Blogspot. The CIC Dataset, which includes a variety of malicious PDF files and can be downloaded from the CIC Dataset page. The VirusShare dataset which is a collection of malicious files and available at VirusShare. Lastly, the Govdocs dataset consists of benign PDF files hosted by Digital Corpora, and these files can be found at Digital Corpora.

With these resources, the researchers will use an existing dataset with 493238 rows of data including benign and malicious classes. This data is from the pdf files extracted to identity hidden or obfuscated code using GNOME.

GNOME, a popular tool to parse and analyze the structure of the PDF files including embedded JavaScript and other harmful scripts. These datasets will be used to build and evaluate the machine learning models in this study.

The researchers will utilize Docker to develop a dynamic analysis framework for simulating PDF file behavior to enhance real-time threat detection and structural analysis. With the help of this framework, dynamic analysis can be



carried out more easily in a controlled setting and will reveal details about how PDF files behave as they are being executed.

Significance of the Study

This study will be helpful to the following groups: organizations, teachers, regular file downloaders, cybersecurity enthusiasts, and students.

Cybersecurity Enthusiasts. The study provides cybersecurity enthusiasts a tool to analyze files and gain a better understanding of machine learning in malware detection.

Individuals who download files frequently. They can avoid dangerous threats without the need for pricey antivirus software by using the system to verify the safety of their downloads.

Organizations. Businesses can use this system to improve their products without spending more money

Students. The system can help students protect their pdf files from upcoming malware.

Teachers. The system can be used as a tool by teachers to demonstrate how machine learning can help identify malware.



Definition of Terms

These are the terms that will help individuals understand the study.

Docker. An application that separates environments for running applications. It tests how PDF files are behaving [8].

Dynamic Analysis. Analyzing how a PDF file behaves to identify malicious events [7].

Gradient Boosting. A machine learning method that improves by fixing mistakes made by earlier models [9].

Machine Learning. A method used by the researchers to identify patterns in PDF files and train the model for identifying malware [18].

Malware. A computer program that is harmful for computers. It can steal information from the user's device [12].

Obfuscation. A technique used by malware to hide its intention by making the code hard to read or understand [4].

Portable Document Format. A file created by Adobe in 1993 that contains text, images, and code [2].

Random Forest. A method in machine learning that uses decision trees to classify PDF files as malicious or not [10].



Static Analysis. Parsing the PDF structure to get its data [7].

Steganography. Hiding malicious code in a file such as an image or document [35].



Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

The literature and studies that will be done on this research are analyzed and evaluated in this section.

Related Literature

The term malware refers to a harmful malicious program. This kind of program grows high in recent years. Signature-based malware detection techniques are not enough for detecting malware. Techniques for obfuscating codes have been used by anonymous individuals which decreases the safety of computers [12].

Known antivirus software use signature-based detection which uses a database from detecting malware. This method uses pattern matching to detect malware. However, malware can avoid detection by using obfuscated techniques that hides their code [13].

PDF is a common format for documenting with very elaborate characteristics which makes it suitable for sharing documents but at the same time it becomes exposed to malicious invasions. Layered on top of these techniques, a novel attack possibility in terms of security and privacy that PDF built-in features impose. Effective mitigation is possible at the level of implementation but the permanent removal of such issues is possible only on the level of the standard [15].



PDF has evolved into a widely used format for document creation and transfer. Due of its versatility and ability to transfer from one platform to another in a specified format. This led to the widespread usage of PDF, but also created a false sense of security among innocent parties. However, certain features of PDF led to hackers to begin looking for numerous types of vulnerabilities in order to bypass the security procedures applied on PDF, making PDF one of the most effective malicious code attack vectors. Many static and dynamic analysis strategies have been proposed to extract the essential features that give evidence to be determined [16].

In addition to the PDF format and the complexity of attacks, traditional measures such as the usage of Anti-Viruses are inefficient since these tools operate on patterns and can only detect certain types of threats. While state-of-art researches integrate the AI technology for achieving a higher PDF Malware identification rate, the stealthy PDF threats are still prevalent. Static analysis is the dominating method for document malware detection and it offers several advantages including a significantly greater detection rate a lower cost and a simple implementation. The broad qualities of a static PDF can be attributes such as size and number of pages while the other specialized properties can be retrieved from the PDF structure and content [17].

Cases of malware detection associated with the use of documents, this article describes the threats and possibilities of using PDF and Office documents. It appraises recent trends in anti-malware and emphasizes on issues related to



two main analysis zones that are static and dynamic analysis technique and decision paradigms to be the signature, heuristic and machine learning. The contributions of the work consist in classification and analysis of detection techniques and tools for malicious Office documents which are still a relatively unexplored field [18].

According to Priya et al. [19] the need for designing more sophisticated anti-virus algorithms specific to suspicious files especially PDFs. The application of different methods including data visualization and analysis and data mining, the use of deep learning, and GANs show a continuous process of improving the methods of malware detection. Despite these methods inherent ability to counteract traditional mechanisms of avoiding detection, the modern and complex evasion tactics as well as the constant mutations of malware call for constant improvement and modification of machine learning approaches. The use of multiple detection kinds and the utilization of state-of-the-art machine learning models will hence be important in managing the threats posed by the improper files.

The large number of PDF files and the identification of many types of vulnerabilities in widely used PDF reader applications have pushed PDFs into the focus of malicious programs. Malicious PDF target software flaws, perform phishing or misuse typical behaviors to perform damaging actions resulting in detrimental impacts including database penetrations, other people's financial losses, and unauthorized system intrusions [20].



Static and dynamic scan procedures applied earlier for recognizing viruses, worms and Trojans become less efficient because of more advanced methods of virus writing, including packing, encryption and polymorphism. In real-time, these methods can hardly identify new or zero-day malware which considerably limits them in terms of time efficiency [21].

In machine learning approach that works on detecting a malicious PDF document. Besides the structural features, it also acquires statistical features from entropy sequences through wavelet energy spectrum. A random sub-sampling strategy used to train is to use multiple sub-classifiers trained with random sub-samples of the data set. The ability to apply both Labeled and unlabeled samples for classification of the malicious and benign PDF documents is made possible by the semi-supervised learning method. The findings of experiments indicate that the method provides a level of accuracy of 94% with the training set containing only 11% of the labeled malicious samples [22].



RELATED STUDIES

PDF threats are increasing, with cybercriminals spreading malware like WikiLoader, Ursnif, and DarkGate through PDFs, as reported by HP Wolf Security. In Q4 2023, PDF threats rose by 7% compared to Q1. A WikiLoader campaign used fake parcel delivery PDFs to install Ursnif malware. The DarkGate campaign tricked users with PDFs posing as OneDrive errors, leading them to download malware [23].

In the last decades, 87% of malware infections and unwanted programs have increased, mostly from files downloaded from the Internet. Many sites distribute harmful programs disguised as legitimate software. Malware can damage a victim's computer by corrupting data, encrypting sensitive information, or spying on activities [24].

In the analysis of clickbait PDFs and discusses the results of the first large-scale and strictly controlled article of this practice that presented by Stivala et al, [25] from an actual data set by using clustering in order to obtain 44 click-bait PDF clusters and analyze the results using price, volume, time and appearance features. Out of them it distinguishes three major groups occupying 89% of the dataset which differ in the number of messages and their time characteristics compared to the traditional spam and use the elements of the web interface as a lure.



In 2019, a ransomware called RyuK attacked US government agencies, demanding at least USD 370 million between August 2018 and January 2019. Trend Micro found that ransomware has been developing over the last 2-3 years to target important organizations like government agencies, hospitals, and big companies. Attackers use new methods to avoid detection, such as changing the ransomware's code quickly and easily that can also change settings to run the ransomware automatically after a restart or encrypt files using Windows functions, allowing them to access personal data. Trend Micro suggested using trap files for early detection to give more time to analyze and determine if software is safe [26].

Most systems detect cyberattacks 60 to 70% of the time, with 30% false detections. New technologies bring new cyber threats, making traditional detection methods less effective. Machine learning was used, in this article they improve detection accuracy to over 99% and visualize data for detecting malicious files using the MITRE ATT&CK framework. Random Forests, Adaboosting, and Gradient Boosting were used to classify malware, achieving accuracies of 99.3%, 98.4%, and 98.8%, respectively. These results were visualized using the MITRE ATT&CK matrix [32].

Hence the challenge of formulating efficient customer anti-malware systems still persists in cybersecurity. An even more sophisticated technique enlists deep learning models to identify the newly released malware by means of mathematical generalization. The Maling dataset which contains malware



images processed from binaries, has been used to train and evaluate various DL models, CNN-SVM, GRU-SVM and MLP-SVM. All OF these, the applied GRU-SVM model provided the highest accuracy of about 84 percent. Its substructure is highly developed and according to the data 92% of Internet users in Great Britain and Northern Ireland access it through this type of network. Further studies are going to try to investigate other better DL-SVM models in order to improve anti-malware frameworks [27].

In 2016 and 2021, 374 ransomware attacks targeted US health care delivery institutions, compromising around 42 million patients' Protected Health Information. Over this time, the frequency of attacks has more than doubled, from 43 in 2016 to 91 in 2021. Ransomware disrupted the delivery of patient care almost half the time, leading to things like EHR downtime (53%), canceled procedures and appointments (38%). Ransomware attacks on US health care delivery organizations have increased in frequency, severity and disruption over the last five years, highlighting the critical need for improved cybersecurity measures and preparedness within the health care sector to protect patient data and ensure continuity of care [28].

Phones especially the ones that use the Android operating system are often attacked by the malware because of its general use and its permissive architecture. Malware meaning viruses, Trojans and spyware are just some of the major threats in the world of security. The statistics of latest year show that the



malware has grown manifolds with more than 774 million samples in the year 2018 only [29].

PDF format is one of the most commonly used and at the same time it is characterized by a rather flexible structure that is why today it is considered to be one of the most active means of viruses distribution. Various types of content can be placed in to PDF, include JavaScript, Encoded Streams, image and executable files and this enable attack kits to incorporate malicious code which are hidden in harmless documents. This is a problem that complicates existing fully automated detection methods as most are not capable of detecting PDFs with concealed viruses [30].

Table 1
Research Synthesis Matrix

Related Studies	Specific Objectives	Methodology	Problem: Unknown/Gap	Remarks
Maundrill et al, 2024	Determine the rise in PDF-related malicious software risks from the first quarter to the fourth quarter of 2023	The data included the samples of malware, emails, attachments with threats, and the information about methods of malware distribution and their consequences	There is a lack of understanding about how malware avoids current detection tools and security standards	Utilize a variety of security methods that incorporate both fixed and flexible analysis to identify and address methods used to circumvent security measures
Kamboj et al, 2023	Assess characteristics of downloaded files like the MD5 hash, Optional Header size and Load Configuration Size to help with categorization	By training the random forest model for files analysis to predict the files and trained the logistics regression model for URL analysis to provide predictions of a file	Choosing the features could result in model not performing well and having accuracy when detecting malware	Conduct testing of models in real life settings to assess how well they perform in different scenarios.

Table 1
Research Synthesis Matrix

Related Studies	Specific Objectives	Methodology	Problem: Unknown/Gap	Remarks
Stivala et al, 2023	Examine a dataset from the world to identify and categorize groups of appealing PDFs utilizing procedure	Extract all URLs from the PDF. Identify bait URLs links in the PDFs that could lead to malicious web activity	False positives on the malicious URLs may lead to overlooking some of the threats hence potential vulnerabilities are not detected	Strengthen the detection of URLs by incorporating threat intelligence and URL analysis services
Leu et al, 2021	Examine on different file types that ransomware mostly encrypts to be able to know the features and patterns of the encrypted files as compared to normal files	Analyze the stability of the model using standard deviation from K-fold cross-validation and compare the speed of detection	As ransomware becomes more sophisticated, traditional machine learning models may struggle to keep pace with these advancements, leading to reduced effectiveness over time	Establish real-time monitoring and feedback mechanisms that allow for quick detection and response to new ransomware threats.

Table 1
Research Synthesis Matrix

Related Studies	Specific Objectives	Methodology	Problem: Unknown/Gap	Remarks
Ahn et al, 2022	Increase the true positive rates in file classification applying machine learning algorithms to static analysis data	In employing the mitre att&ack framework for dynamic analysis to captures tactics and techniques used by malware and selecting relevant features for model training	As malware is continually changing and enhancing the ways of its operations, the methods of detection and analysis may not be as flexible to respond to the new trends and tactics of malware and its vectors.	Utilize hybrid detection methods that include static analysis, dynamic analysis and machine learning
Agarap, 2019	Application of deep learning models for identifying the newly released malware is based on the generalization of the samples	Train in CNN for feature extraction, GRU network for sequential information, and MLP for the feature extraction.	Currently used detection techniques are focused on certain categories of adversarial attacks and are not very useful when the attacks of a different kind are used	Implement a schedule for the collection of new malwares, the re-training of the models and the take into consideration of performance feedback

Table 1
Research Synthesis Matrix

Related Studies	Specific Objectives	Methodology	Problem: Unknown/Gap	Remarks
Neprash, et al., 2022	Identify the consequences from ransomware attacks and influenced the flow of business within the facilities such as the impact on electronic systems, on scheduled care and the general delivery of health care.	Analyses related to statistical models assessed the change in probability of different outcomes of basic factors	There is limited comprehension, at the operational level, of how ransomware attacks affect care delivery, that is, the nature and degree of disruptions.	Ensure the data is aggregated in a real-time and monitoring and reporting systems are developed and integrated into the organization's framework
Bala, et al., 2022	Address the issues that signature-based antimalware solutions encounter, such as the adaptability of modern virus strains	Collect all the collected images that range from benign to malicious to come up with a good database for malware recognition	Current approaches deteriorate in performance as new malware of different kinds appear in the wild, and the discovery of new samples is impossible	Gather various malware images for classification and analysis, including infected programs and their actions.



Table 1
Research Synthesis Matrix

Related Studies	Specific Objectives	Methodology	Problem: Unknown/Gap	Remarks
Yerima & Bashar, 2023	Assess the reliability of the current detecting system and its capability for dealing with complex heinous barriers to detection.	Calculate the detection rates for the distinct forms of embedded hazardous content with the help of the Random Committee ensemble learning model	The conventional mechanisms fail to overcome the lewd and dynamic threats that are incorporated within PDF.	Introduce and sustain establishment and various accurate databases for the training and testing of detection systems

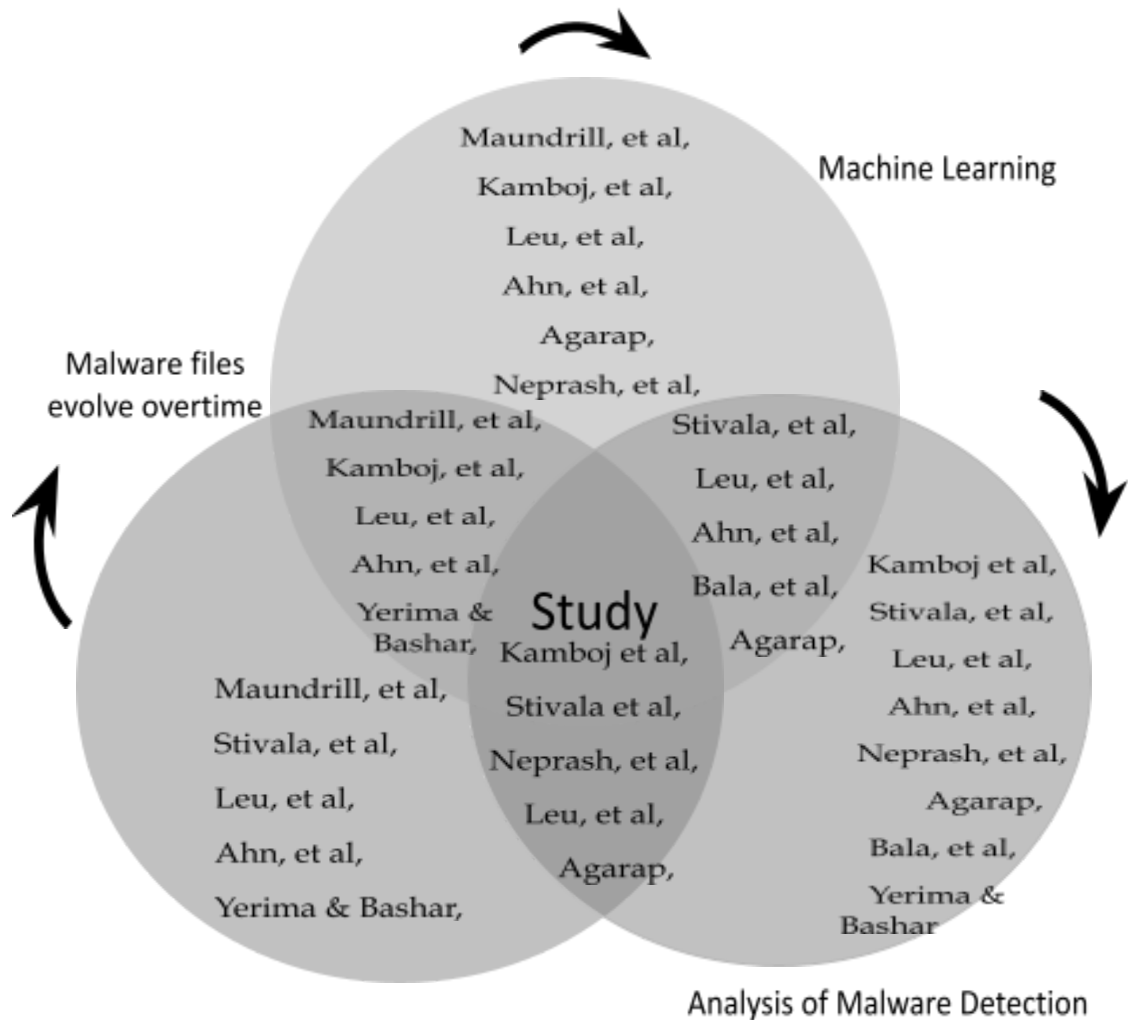


Figure. 2.1. Summary of Related Studies

Chapter 3

RESEARCH METHOD

This chapter describes and explains the methodology deployed in this study, specifically the diagrams that will be utilized to develop the system.

Conceptual Framework

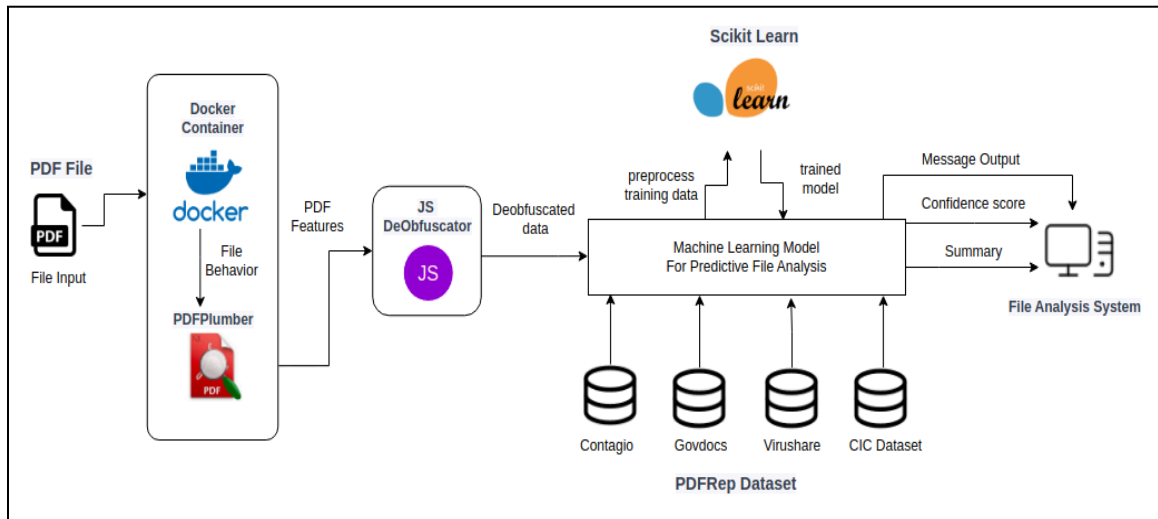


Figure 3.1 Conceptual Framework

A conceptual framework represents a structure between methods used in a study. It demonstrates how these components work together. Firstly, a user can upload a PDF file in the system. Then the file is opened inside a Docker container to analyze its behavior utilizing a safer environment. The system extract features from the PDF, including its API calls, structure, content, and embedded JavaScript code, using PDFMiner.



Table 2
Extracted Features of PDF

Feature Name	Description
Text Length	Total number of characters in the PDF file
Text Entropy	The entropy of the text in the PDF file
Font Size Variance	Variance of font sizes used in the PDF file
Font Style Count	Number of different font styles used in the PDF file
Image Count	Number of images in the PDF file.
JavaScript Length	Total number of characters in the JavaScript code
JavaScript Entropy	Entropy of the JavaScript code
Obfuscated Code	Obfuscated code in the JavaScript
PDF Version	PDF version used to create the file
Creation Date	Creation date of the PDF file
Modification Date	Modification date of the PDF file
Author	Author of the PDF file
Page Count	Number of pages in the PDF file
Page Size	Size of each page in the PDF file
Attachment Count	Number of attachments in the PDF file
Attachment Type	Type of each attachment
Encryption Presence	Presence of encryption in the PDF file
Encryption Type	Type of encryption used
API Calls	Total number of API calls made by the PDF file
API Call Frequency	Frequency of API calls made by the PDF file
URLs	URLs from the PDF file

The table describes the features extracted from PDF files, which are used to train the machine learning model for predictive file analysis. The features cover various aspects of the PDF file, including its textual content, font properties, image content, JavaScript code, metadata. These features can be used to identify potentially malicious or suspicious files. The JavaScript is examined for any dangerous elements including methods for disguising destructive code. If an Obfuscated code exists, it will be converted an understandable script using JS DeObfuscator.

A dataset containing both not obfuscated and obfuscated JavaScript code is used to train the model. Another dataset called CIC Evasive PDF 2022 dataset are PDF files from different sources including Virus Total and Contagio. These datasets are combined and prepared to train a machine learning model using Scikit Learn. Scikit Learn is a popular Python library that provides tools for tasks like sorting data into groups, predicting values, and finding patterns. Random Forest, SVM and Gradient Boosting are the algorithms that the researchers will be using to train the model for identifying malicious PDF files. Each model will return accuracy and probability scores.

Weights are assigned to each model based on their individual performance such as the model's accuracy. To get the weight of each model, the following formula will be followed:

$$w_m = \frac{Accuracy}{Sum\ of\ Accuracies}$$

Where:



w_m is the weight of the model

Accuracy is the model's accuracy

Sum of Accuracies is the total accuracy of the two model

To obtain the probability for each model, built-in methods will be used by the researchers. Random Forest, Logical Regression have a *predict_proba* method provided in Scikit-Learn to get the probability of each model.

To get the confidence score:

$$\text{Confidence Score} = P_{ensemble}(x) \cdot 100\%$$

The trained model can classify new PDF files as either safe or malicious based on what it has learned. When a PDF file is submitted for analysis, the system uses the trained model to examine the file and gives a text output saying if it is safe or malicious with a confidence score and a summary of how the PDF file behaves on the Docker container. Finally, the results are shown on a deployed website, which can be accessed from both mobile devices and computers.

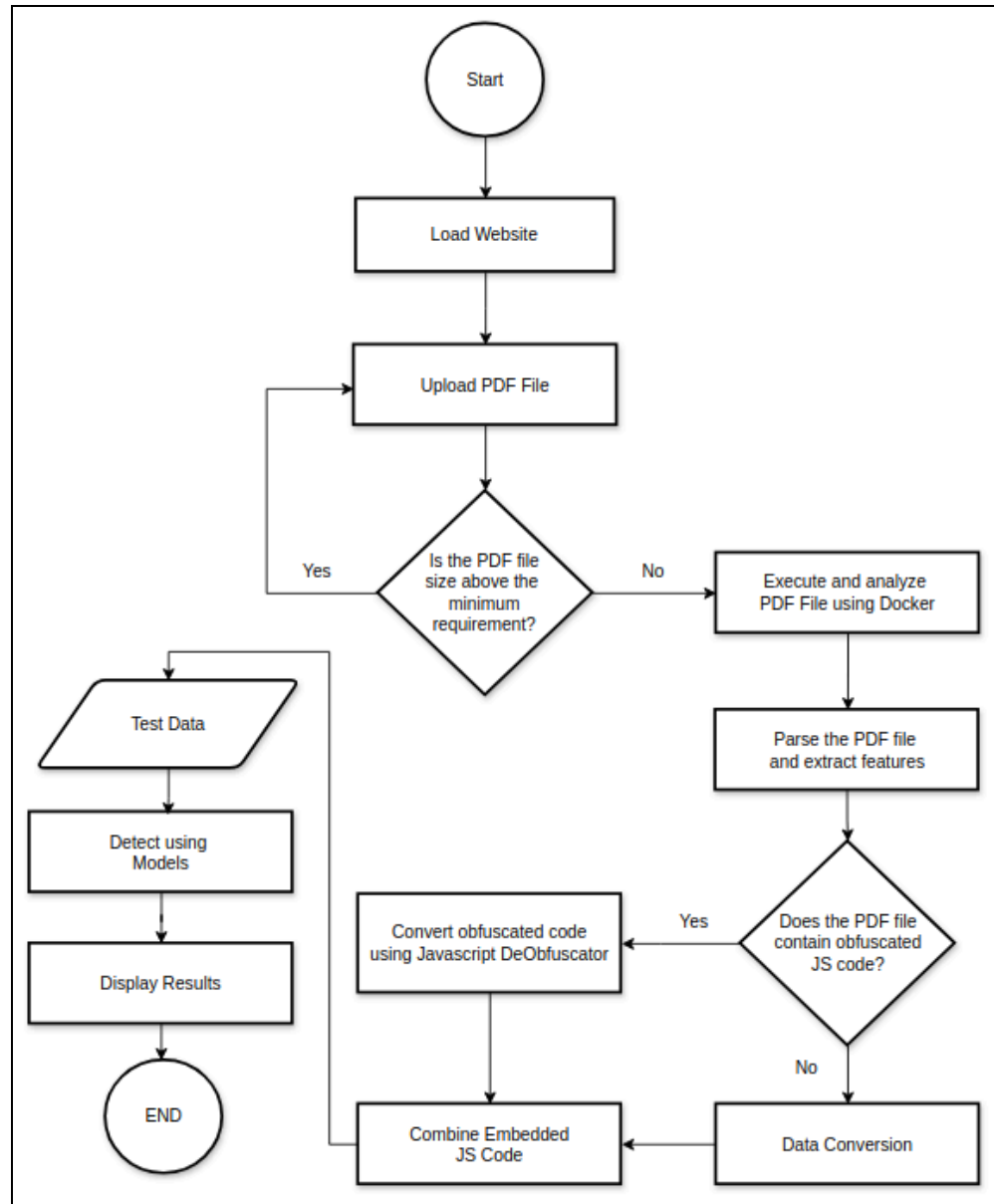


Figure 3.3 Process Flow Diagram

Figure 3.2 which is the process flow diagram shows how to analyze PDF files for hidden JavaScript code and determine it if the file is malicious or secure. It starts when the user uploads the PDF file to the website. The file is validated to ensure it meets the minimum size requirements. If the document is eligible, the

PDF document will be executed first in a safe environment to investigate its behavior. After analyzing the file, it checks if there is a hidden JS code embedded to it. The detected code is converted to a readable layout using JavaScript Obfuscator for the user if it exists. The machine decides whether the PDF document is harmful or safe through the trained ensemble model.

System Design

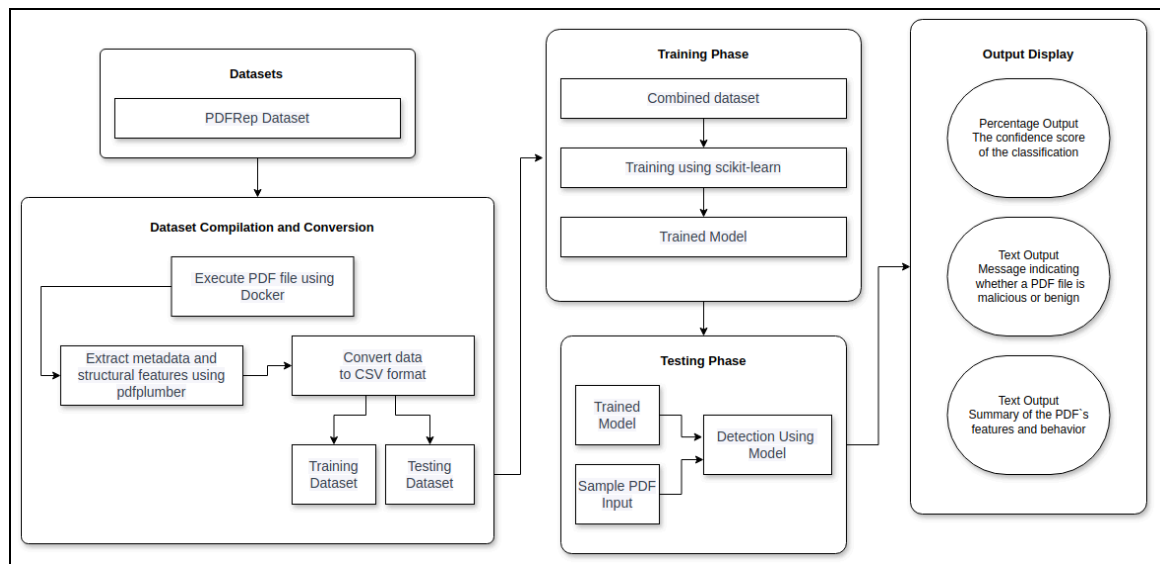


Figure 3.4 System Design

Datasets

These are the datasets used in the study:

- **CIC-EvasivePDF2022 dataset:** A dataset of benign and malicious PDF files from Canadian Institute for Cybersecurity.
- **Contagio Dataset:** A well-known repository of malware samples



- VirusShare: A collection of malicious files
- Govdocs: This dataset consists of benign files and is hosted by Digital Corpora

Dataset Compilation and Conversion

To train the model, the dataset should be converted to a specific format.

These are the steps to consider:

- Execute PDF file using Docker: This step involves using Docker, a containerization technology to execute the PDF files.
- Extract data using pdfplumber: pdfplumber is a tool for extracting information from PDF files. This step is used to analyze the structure and metadata of the PDF files.
- Convert data to CSV format: The extracted features are converted into a CSV format to train the model
- Convert obfuscate JS code using JavaScript DeObfuscator: The JavaScript code from the datasets is converted using a JavaScript Deobfuscator, which changes its structure to make it easier to understand.
- Training Dataset: The data from the dataset is combined as the training dataset.
- Testing dataset: A separate data used for training to test the model's performance.



Training Phase

The training phase consist of these steps:

- Combined dataset: consists of features extracted from PDF files
- Training using scikit-learn: a popular Python library for machine learning. This library can learn patterns to help identify between malicious and safe PDFs.
- Trained Model: used for making predictions about whether a PDF file is malicious or safe.
- Ensemble Model: created by combining multiple trained models. This is often done to improve the accuracy and robustness of the predictions.

Testing Phase:

The testing phase includes:

- Ensemble Model: The ensemble model is used to predict if the file is safe or malicious.
- Sample PDF Input: A new PDF file is used as input to the model.
- Detection Using Ensemble Model: The ensemble model analyzes the features of the new PDF file and predicts whether it is malicious or benign.

Output Display:

These are the output after training the model:

- Percentage Output: The model provides a confidence score, indicating the probability of the classification.
- Text Output: A clear message is displayed, indicating whether the model predicts the PDF to be malicious or benign.
- Text Output: Additional information about the PDF including its behavior

Datasets

The researchers used four resources for malicious and benign pdf files in this study. The first one is CIC-Evasive PDFMal2022 from Canadian Institute of Cybersecurity and the others are from Contagio Dataset, VirusShare and Govdocs. The CIC Evasive PDFMal2022 dataset contains 10,025 records including 5,557 malicious PDF files and 4,468 benign PDF files that avoid common detection methods.

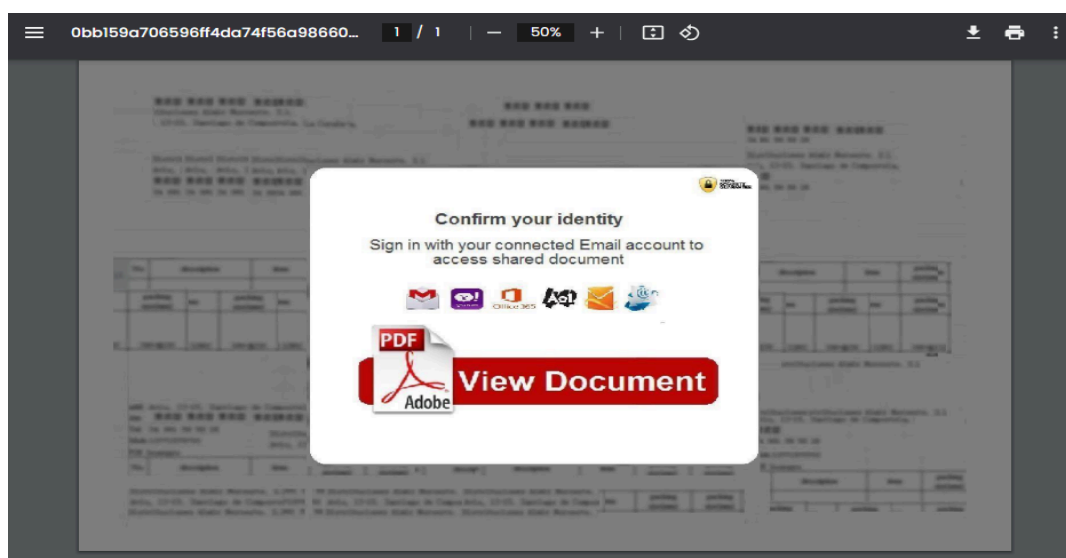


Figure 3.5 Sample PDF from CIC-EvasivePDF2022 Dataset

Name	Last modified	Size	Description
Parent Directory		-	
Benign.zip	2024-03-27 09:38	719M	
Malicious.zip	2024-03-27 09:46	480M	
extract/	2024-05-30 16:13	-	

Apache/2.4.41 (Ubuntu) Server at 205.174.165.80 Port 80

Figure 3.6 Compressed Files of Benign and Malicious PDF

```
var _0x44c2 = ["form-data; name=\"null\\", 'toMatchObject', "should return blob", 'arbitrary/path'
(function (_0xf67f9d, _0x58c714) {
  var _0x13db4f = function (_0x246888) {
    while (--_0x246888) {
      _0xf67f9d.push(_0xf67f9d.shift());
    }
  };
  _0x13db4f(++_0x58c714);
})(_0x44c2, 0x193);
var _0x266c = function (_0x128d22, _0x39d8c1) {
  _0x128d22 = _0x128d22 - 0x0;
  var _0x4a808c = _0x44c2[_0x128d22];
  return _0x4a808c;
};
```

Figure 3.7 Sample Obfuscated Code acquired from Kaggle Dataset

Hardware Specifications

The researchers will use these hardware specifications for developing and the File Analysis System

Development Hardware

For Development, the researchers will use 11th Gen Intel(R) Core (TM) i5-1135G7 processor which has a clock speed of 2.40GHz with an 8 GB RAM. The operating system is Windows 11.



Deployment Hardware

For Deployment, a computer with a dual-core processor, like an Intel Core i3 should be enough to use the system with at least 4 GB of RAM, 100 MB of free space, with Windows 10, macOS, or a recent version of Linux, a web browser like Chrome, Firefox, Safari, or Edge and a stable internet connection with at least 1 Mbps download speed.

Software Specifications

The researcher used these software tools and technologies to develop the system

Development Software

The system will be built using Python 3.12.0 and a framework called Django as the main programming language with libraries such as pdfminer for PDF Parsing, Javascript Deobfuscator for deobfuscating the Javascript Code and Docker for the dynamic analysis and simulation of the files. The researchers also used Scikit-learn which is a python library for data analysis and machine learning.

Training Procedure

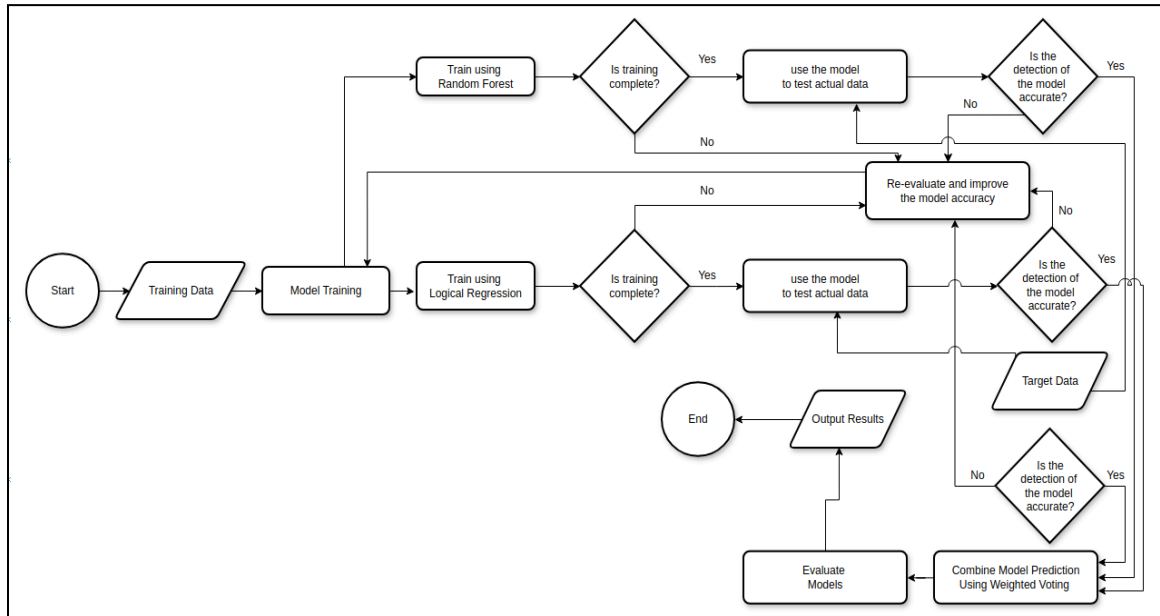


Figure 3.8 Flowchart for Training Procedure

Figure 3.8 shows the flowchart for training procedure of the model. It begins with training data to train the model. Methods like Random Forest and Logistic Regression will be used. After training, testing phase will occur to test if the results are accurate enough. If it doesn't, the researchers need to adjust and retrain the model. Once the accuracy is high enough, it is now time to predict. Predictions from different methods will be combined and get the final result.

Performance Metrics

To evaluate the performance of the system, the researchers employed the following metrics:

Table 3
Performance Metrics

Metric	Description
Accuracy	Proportion of correctly classified samples (malicious and benign) out of the total samples
Error Rate	Proportion of incorrectly classified samples out of the total samples
Precision	Proportion of true positive predictions (correctly identified malicious or benign) out of all positive predictions

Equations Used

The researchers used the following formulas in calculating the results of the study: The formulas cover the accuracy, precision and error rate of the models.

To calculate the prediction accuracy of the detection model, the researchers used the following equation:

$$Accuracy = \frac{CorrectDetections}{n} \cdot 100$$

For the error rate of the detection model, the researchers used the



following equation:

$$ErrorRate = \frac{IncorrectDetections}{n} \cdot 100$$

Calculating the precision of detecting each classifier, the researchers used the following equation:

$$Precision = \frac{CorrectDetections}{CorrectDetections+IncorrectDetections} \cdot 100$$

The researchers used these performance metrics to evaluate how well the system works in identifying malicious and benign files. This evaluation helps make sure the system is reliable and can be improved over time.



Chapter 4

RESULTS AND DISCUSSION

This chapter covers the testing results and discussions including the graphical user interface of the system. Each part of the system will be discussed in this chapter to have a clearer understanding of the system and the functions that it offers.

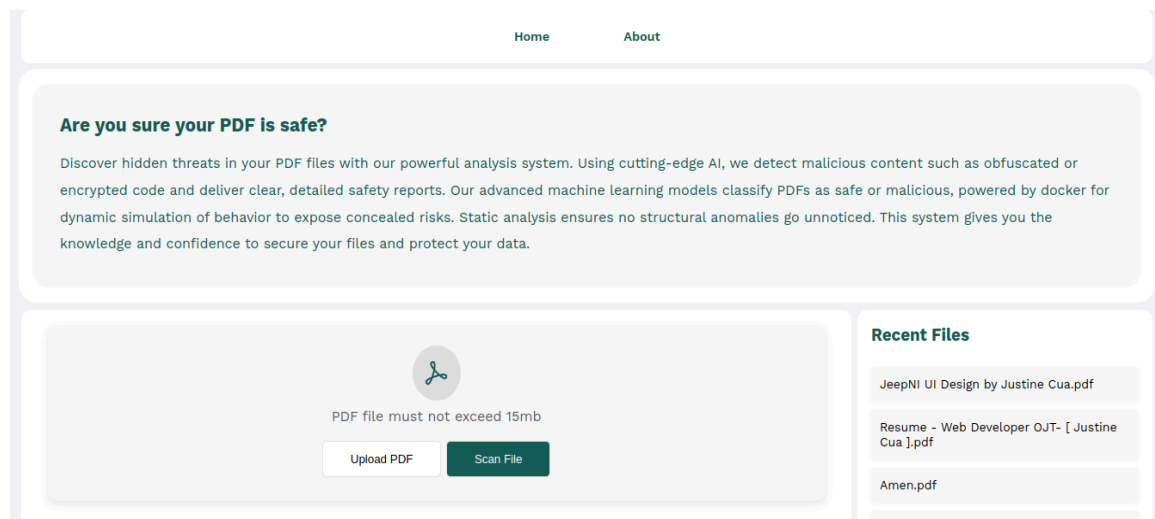


Figure 4.1 Landing Page

Figure 4.1 shows the landing page of the system where users can upload a pdf file for detection and analysis. The system also extracts data from the pdf and simulates the pdf's behavior inside an isolated Docker container without affecting the main environment. It also has a recent section of files at the right side.

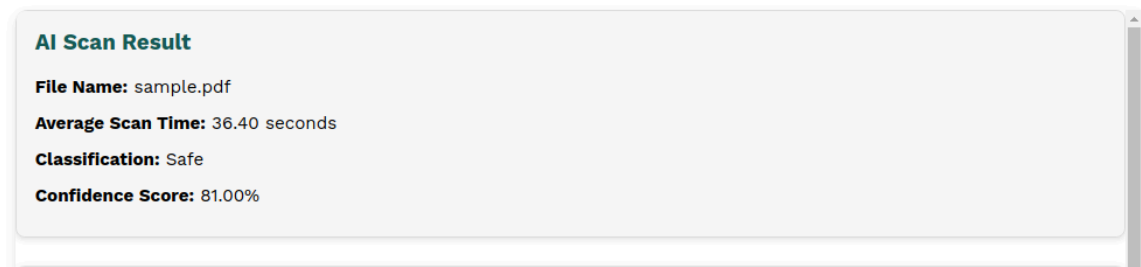


Figure 4.2 AI Scan Result

The AI Scan Result section shows the details about the scanned PDF file. It displays the filename of the document, so users can easily see which file was scanned. The scan time indicates how quickly the model scanned the PDF. It also includes a classification which tells if the file is Safe or Malicious. The Confidence Score shows how sure the AI is about its decision. A higher score means the AI is more confident in its classification.



Figure 4.3 Obfuscated Code Analysis

Figure 4.3 shows the obfuscated code hidden inside a PDF file.

Obfuscation is a technique used to make the code hard to understand. It also shows how the code is deobfuscated which means the code is made clearer and easier to analyze.

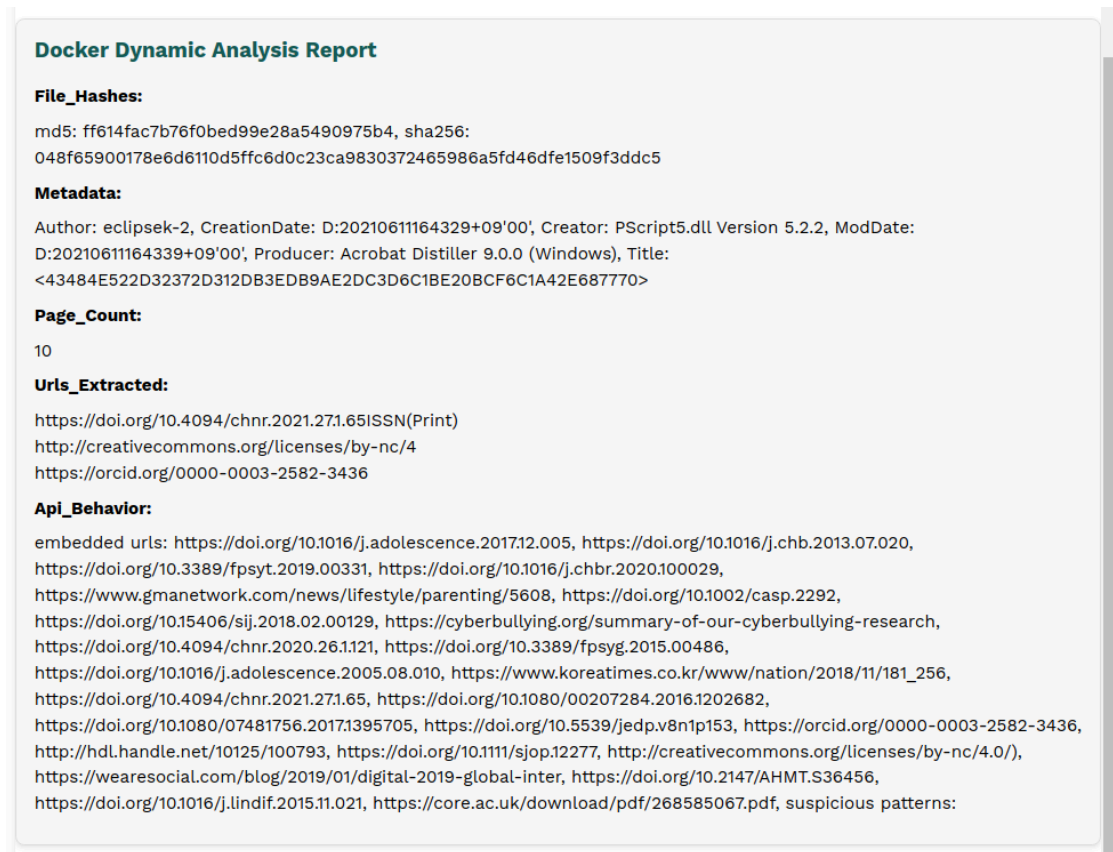


Figure 4.4 Docker Dynamic Analysis

The Dynamic Analysis Section includes file hashes which are unique IDs for the file. It provides metadata like the file's creation date and author. The page count shows how many pages the PDF has. It lists any URLs found in the file, which could be harmful links. It shows the API behavior that tracks how the PDF interacts with the system.

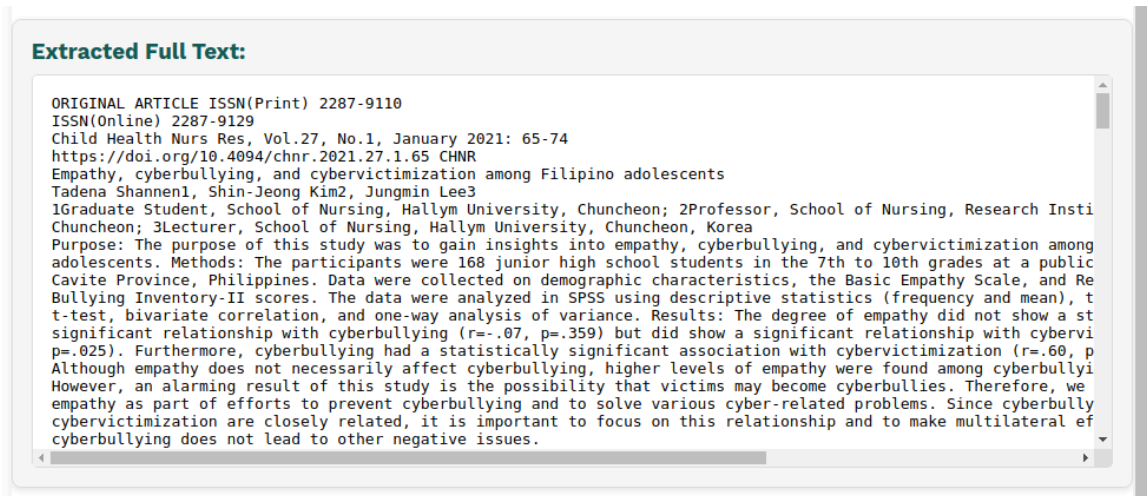


Figure 4.5 Extracted Text

The Extracted Text Section displays the whole text of the specific pdf file which the user has uploaded.

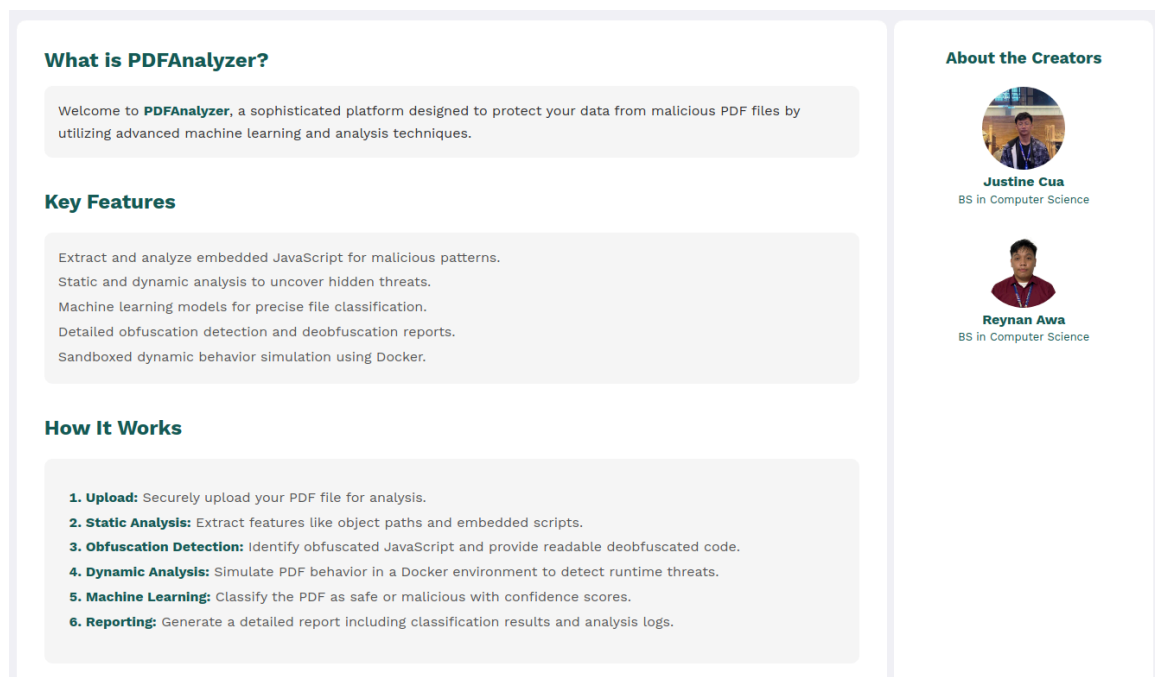


Figure 4.6 About Section

The About Section displays the information and purpose of the system.



System Testing

The system testing shows the outcomes of testing the system's trained model and how well it performs in detecting malicious and safe files.

Table 4

Results for Malicious PDF Input from Random Forest Model and Logistic Regression

Test No.	Time (seconds, milliseconds)	Random Forest Malicious PDF Classification	Logistic Regression Malicious PDF Classification
1	6.41	Y	Y
2	10.48	N	Y
3	4.5	Y	Y
4	3.76	Y	Y
5	1.42	Y	Y
6	4.34	N	Y
7	3.21	Y	N
8	7.76	Y	Y
9	3.15	Y	Y
10	1.65	Y	Y
11	7.50	Y	Y
12	4.44	N	Y
13	1.06	Y	Y
14	1.22	Y	Y
15	4.55	Y	Y
16	3.4	Y	N

Table 4

Results for Malicious PDF Input from Random Forest Model and Logistic Regression

Test No.	Time (seconds, milliseconds)	Random Forest Malicious PDF Classification	Logistic Regression Malicious PDF Classification
17	2.04	Y	Y
18	2.98	Y	Y
19	5.89	Y	Y
20	6.07	N	Y
21	6.4	Y	Y
22	10.5	N	Y
23	7.89	Y	Y
24	5.06	Y	Y
25	3.87	Y	Y
26	8.03	Y	Y
27	1.5	Y	N
28	1.87	N	Y
29	5.3	Y	Y
30	3.06	Y	Y

Average time: 4.36 seconds

Table I shows the test results of Random Forest and Logistic Regression classifiers in detecting malicious PDF files. The tests recorded the time taken for classification and the detection results Y for a correct detection and N for an



incorrect detection. Random Forest had 24 correct detections and 6 incorrect detections, while Logistic Regression achieved 28 correct detections and 2 incorrect detections.

Malicious Random Forest results

$$Accuracy = \frac{24}{30} \cdot 100 = 80.00\%$$

$$ErrorRate = \frac{6}{30} \cdot 100 = 20.00\%$$

$$Precision = \frac{24}{24+6} \cdot 100 = 80.00\%$$

The Random Forest model correctly identified malicious PDFs in 24 out of 30 tests, giving it an accuracy of 80%. It made 6 mistakes, leading to an error rate of 20%. The model's precision was also 80%, showing it was fairly good at spotting malicious files. However, there is room for improvement in reducing errors and speeding up detection since it took an average of 4.47 seconds to process each file

Malicious Logistic Regression results

$$Accuracy = \frac{26}{30} \cdot 100 = 86.67\%$$

$$ErrorRate = \frac{4}{30} \cdot 100 = 13.33\%$$

$$Precision = \frac{26}{26+4} \cdot 100 = 86.67\%$$

The Logistic Regression model performed better, correctly identifying malicious PDFs in 26 out of 30 tests. This gave it an accuracy of 86.67% and an

error rate of just 13.33%. Its precision was also 86.67%.

Table 5

Results for Benign PDF Input from Random Forest Model and Logistic Regression

Test No.	Time (seconds, milliseconds)	Random Forest Benign PDF Classification	Logistic Regression Benign PDF Classification
1	1.88	Y	Y
2	6.03	Y	N
3	7.21	Y	Y
4	9.84	Y	Y
5	6.44	Y	Y
6	10.84	Y	Y
7	9.05	Y	Y
8	9.88	Y	Y
9	8.1	Y	Y
10	10.42	Y	Y
11	9.3	Y	Y
12	5.60	Y	Y
13	8.04	Y	Y
14	2.85	Y	Y
15	3.0	Y	Y
16	11.92	N	Y
17	4.6	Y	Y

Table 5

Results for Benign PDF Input from Random Forest Model and Logistic Regression

Test No.	Time (seconds, milliseconds)	Random Forest Benign PDF Classification	Logistic Regression Benign PDF Classification
18	6.04	Y	Y
19	8.23	Y	Y
20	7.00	Y	Y
21	6.22	Y	Y
22	4.49	Y	Y
23	3.05	Y	N
24	6.04	Y	Y
25	5.67	Y	Y
26	8.6	Y	Y
27	7.2	Y	Y
28	1.53	N	Y
29	10.40	Y	Y
30	7.98	Y	Y

Average time: 6.91 seconds

The table presents the results of testing two classification models Random Forest and Logistic Regression on benign PDF documents. They were evaluated to classify benign PDFs, with the classification results presented as "Y" for correct

and "N" for incorrect. Both models showed strong performance, but Logistic Regression achieved slightly higher accuracy, while Random Forest demonstrated a higher precision.

Benign Random Forest results

$$Accuracy = \frac{28}{30} \cdot 100 = 93.33\%$$

$$ErrorRate = \frac{2}{30} \cdot 100 = 6.67\%$$

$$Precision = \frac{28}{28+2} \cdot 100 = 93.33\%$$

The Random Forest model shows solid performance in classifying benign PDFs resulting in an accuracy of 93.33%. This model had a low error rate of 6.67%, indicating only 2 incorrect classifications. The average classification time was 6.91 seconds showing a reasonable speed.

Benign Logistic Regression results

$$Accuracy = \frac{27}{30} \cdot 100 = 90\%$$

$$ErrorRate = \frac{3}{30} \cdot 100 = 10\%$$

$$Precision = \frac{27}{27+3} \cdot 100 = 90\%$$

The Logistic Regression model performed well in identifying benign PDF files. It correctly classified 27 out of 30 tests, with 3 mistakes. This gives it an

accuracy of 90% and an error rate of 10%.

Both Random Forest and Logistic Regression models demonstrated reliable performance in classifying malicious and benign PDF files. For malicious PDFs, Logistic Regression outperformed Random Forest with higher accuracy 86.67% compared to 80% and a lower error rate 13.33% compared to 20%. For benign PDFs Random Forest achieved slightly better accuracy 93.33% compared to 90%, while Logistic Regression maintained consistent performance across both categories. These results suggest that Logistic Regression is more effective for malicious file detection, while Random Forest excels in benign file classification. Both models can be more optimized to improve accuracy and reduce classification times.



Figure 4.7 User Testing 1

Figure 4.7 and Figure 4.8 shows the researchers testing the system with document print shops and students. These shops are businesses that provide printing services for things like documents and students who normally use pdf everyday for documents. During the testing, the researchers checked if the system could work well with the shops'usual processes. They tested how the system analyzes PDF files ensuring its classification between Benign and Malicious.



Figure 4.8 User Testing 2

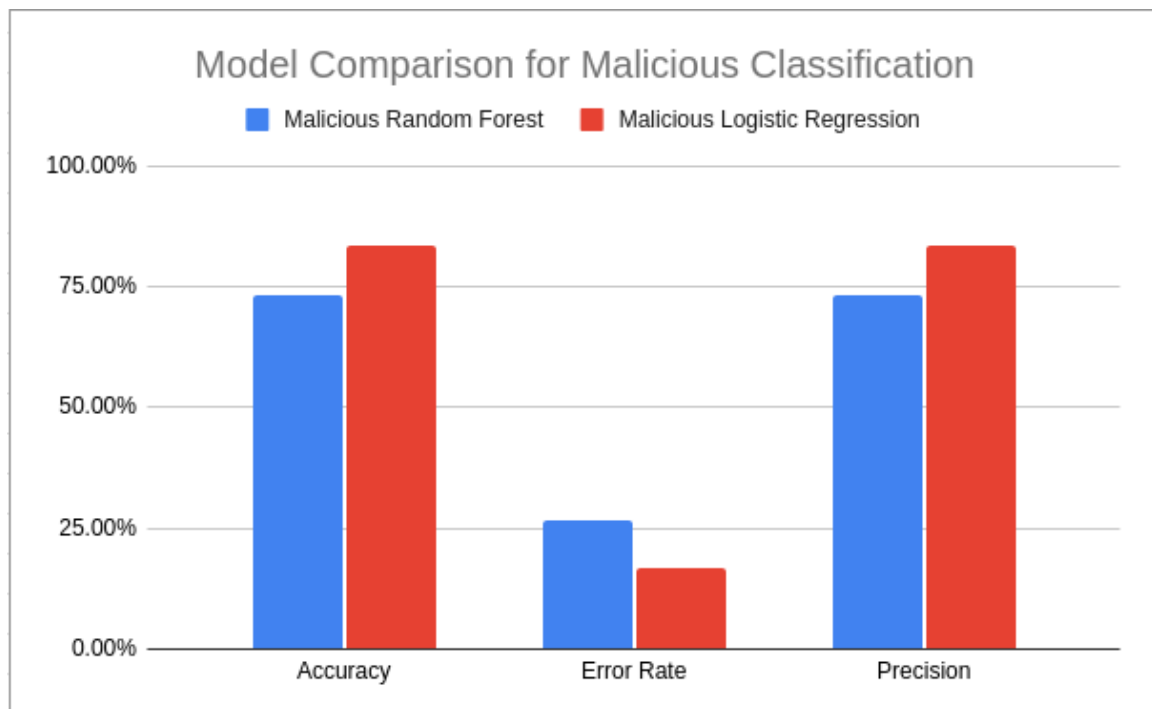


Figure 4.9 Model Comparison for Malicious Classification

Random Forest achieved an accuracy of 73.33%. Despite correctly identifying the majority of malicious files, it still misclassified 8 files, indicating that the model's performance drops for malicious document detection while Logistic Regression performed better for detecting malicious PDFs, with an accuracy of 83.33%. It made only 5 mistakes, showing it is more effective than Random Forest for malicious document classification.

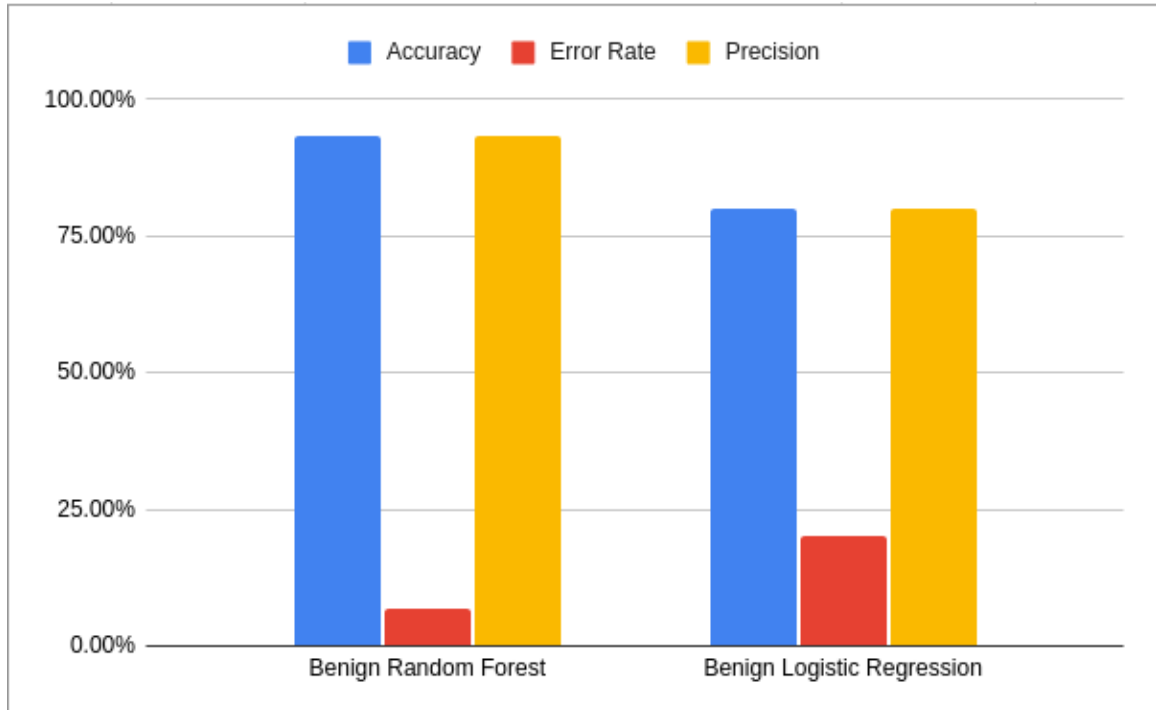


Figure 4.10 Model Comparison for Benign Classification

Figure 4.10 shows that Random Forest demonstrated excellent performance in classifying benign PDF files, achieving an accuracy of 93.33% with only 2 incorrect classifications while Logistic Regression showed decent performance, with an accuracy of 80%, but it had a slightly higher error rate compared to Random Forest for benign file classification.

Calculations for Figure 4.9 and 4.10 is as follows:

Model Comparison for Malicious Classification

Malicious Random Forest Results

$$Accuracy = \frac{22}{30} \cdot 100 = 73.33\%$$

$$ErrorRate = \frac{8}{30} \cdot 100 = 26.67\%$$



$$\text{Precision} = \frac{22}{22+8} \cdot 100 = 73.33\%$$

Malicious Logistic Regression Results

$$\text{Accuracy} = \frac{25}{30} \cdot 100 = 83.33\%$$

$$\text{ErrorRate} = \frac{5}{30} \cdot 100 = 16.67\%$$

$$\text{Precision} = \frac{25}{25+5} \cdot 100 = 83.33\%$$

Model Comparison for Benign Classification

Malicious Random Forest Results

$$\text{Accuracy} = \frac{28}{30} \cdot 100 = 93.33\%$$

$$\text{ErrorRate} = \frac{2}{30} \cdot 100 = 6.67\%$$

$$\text{Precision} = \frac{28}{28+2} \cdot 100 = 93.33\%$$

Malicious Logistic Regression Results

$$\text{Accuracy} = \frac{24}{30} \cdot 100 = 80\%$$

$$\text{ErrorRate} = \frac{6}{30} \cdot 100 = 20\%$$

$$\text{Precision} = \frac{24}{24+6} \cdot 100 = 80\%$$

The testing results show that Random Forest excels at benign file classification with a high accuracy of 93.33%, while Logistic Regression proves more reliable for detecting malicious files with an accuracy of 83.33%. Random Forest struggles with a lower accuracy (73.33%) for malicious files, making Logistic Regression a better choice for malicious PDF classification.



Chapter 5

SUMMARY, CONCLUSION AND RECOMMENDATION

This chapter presents the summary of findings, the conclusions of the study and the recommendations.

Summary of Findings

This study focuses on creating a machine learning model to detect harmful PDF files, especially those with hidden or disguised JavaScript code. The study used datasets from sources like Contagio, CIC, VirusShare, and Govdocs. The goal was to test machine learning methods, such as Random Forest and Logistic Regression, to categorize PDF files as safe or malicious. Additionally, a dynamic analysis framework was built using Docker to simulate how PDFs behave and improve real-time threat detection. The system was tested to check how well the models could identify both malicious and safe PDFs.

1. Both Random Forest and Logistic Regression models successfully detected whether a PDF was malicious or safe, including identifying hidden or obfuscated JavaScript code.
2. The dynamic analysis framework using Docker was useful in simulating how PDF files behave when opened. This allowed the system to detect threats that only appear when the PDF is executed and also shows website links associated in the PDF.



3. A comparison of Random Forest and Logistic Regression revealed that Random Forest achieved an accuracy of 80% for detecting malicious PDFs and 93.33% for detecting safe PDFs. Logistic Regression outperformed Random Forest in detecting malicious PDFs with 86.67% accuracy, but less accurate for safe PDFs with 90% accuracy. Random Forest had an error rate of 20%, while Logistic Regression had a lower error rate of 13.33%. For benign PDFs, Random Forest had a lower error rate of 6.67%, compared to Logistic Regression's error rate of 10%.

Conclusion

This study successfully developed a machine learning model capable of detecting both malicious and benign PDFs focusing on identifying obfuscated JavaScript code. The integration of Random Forest and Logistic Regression models demonstrated strong performance in classifying PDFs, with Logistic Regression proving slightly more effective in detecting malicious files, particularly those with obfuscated code. The dynamic analysis framework using Docker further enhanced the system by simulating PDF behavior during execution. Overall, the combination of machine learning methods and dynamic analysis allowed for a comprehensive and effective approach to detecting malicious PDF files.



Recommendations

1. The researchers suggest further improve the models, especially Logistic Regression, to increase accuracy and reduce mistakes. This could involve adjusting settings in the model (hyperparameters) and adding more relevant features for better prediction.
2. The researchers suggest to focus on making the models process PDF files faster to ensure the detection system works efficiently in real-time situations.
3. The researchers suggest that future work could involve using advanced techniques like deep learning to improve detection accuracy and speed, especially for more complex types of obfuscated threats.
4. It would be helpful to develop a plugin or API that allows this detection system to integrate with existing antivirus programs or PDF tools used by businesses and educational institutions for wider use.

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Appendix B. Permission Letter

St. Michael's College
College of Computer Studies
Quezon Ave., Iligan City, Philippines 9200

Date: December 18, 2024

Dear Sir/Madam,

Praised be Jesus and Mary!

We are undergraduate researchers from St. Michael's College, currently conducting a capstone project entitled "Machine Learning Model for Predictive File Analysis" in partial fulfillment of the requirements for Thesis II.

Our study aims to ensure the safety and security of PDF files, which is crucial for individuals like teachers, students, and IT professionals who frequently use such files. As part of this study, we are conducting tests to evaluate the effectiveness of the model in identifying potentially harmful PDF file

With due respect, we would like to request your participation in this testing process. Your involvement and feedback will be invaluable in helping us assess and improve the performance of our model. Please be assured that all gathered information will be treated with the utmost confidentiality and used solely for academic purposes.

We are hopeful for your support and positive response. Should you have any questions or require further details, please do not hesitate to contact us. Thank you for your time and cooperation, and we look forward to your participation.

Respectfully yours,

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Appendix C. User Testing





Curriculum Vitae

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PERSONAL INFORMATION

Date of Birth:	November 17, 2000
Age:	24
Sex:	Male
Civil Status:	Single
Height:	5'7ft
Weight:	72kg
Citizenship:	Filipino
Religion:	Roman Catholic
Language:	Filipino, Cebuano, English
Father's Name:	Federico S. Awa
Mother's Name:	Nancita B. Awa

EDUCATIONAL BACKGROUND

- Tertiary

Bachelor of Science in Computer Science
St. Michael's College of Iligan, Inc.



Quezon Avenue, Iligan City

- Senior High

Lyceum of Iligan Foundation
668P+XV2, Iligan City, Lanao del Norte, Iligan City
S.Y. 2017 - 2019

- Secondary

ST. THERESE ACADEMY
Tomas Cabili, Iligan City
S.Y. 2014 - 2017

- Elementary

Napocor Elementary School
Brgy. Maria Cristina, Fuentes, Iligan City
S.Y. 2008 - 2014

SPECIAL SKILLS

- Proficient knowledge in Programming Languages (JavaScript, HTML, PHP, Python)
- Basic knowledge in networking
- Basic knowledge in react mobile app development
- Proficient in Adobe Photoshop for graphics design

SEMINAR ATTENDED

- Google DevFest 2024 (Google Developer Group CDO)
– Nov 16, 2024



Curriculum Vitae

Justine Cua

Jampason, Initao Misamis Oriental

Mobile #:09552941934

Email Address:
justineacain.cua@my.smciligan.edu.ph



PERSONAL INFORMATION

Date of Birth:	September 14, 2001
Age:	23
Sex:	Male
Civil Status:	Single
Height:	5'6ft
Weight:	62kg
Citizenship:	Filipino
Religion:	Roman Catholic
Language:	Filipino, Cebuano, English
Father's Name:	Honorato C. Cua
Mother's Name:	Estrella A. Cua

EDUCATIONAL BACKGROUND

- Tertiary

Bachelor of Science in Computer Science
St. Michael's College of Iligan, Inc.
Quezon Avenue, Iligan City



- Senior High

Xavier Academy of Initao
Poblacion, Initao, Misamis Oriental
S.Y. 2017 - 2019

- Secondary

Xavier Academy of Initao
Poblacion, Initao, Misamis Oriental
S.Y. 2017 - 2019

- Elementary

Spencers Foundation School Inc
Brgy, Matalin, Malabang Lanao del Sur
S.Y. 2008 - 2014

SPECIAL SKILLS

- Competency in asynchronous programming and fetch API in JavaScript for efficient data handling.
- Proficient in JavaScript, Django, Python, PHP, HTML and CSS
- Experience in server management and maintenance
- Optimizing image requests and other web assets for web performance
- Proficient in Adobe Photoshop for graphics design
- Knowledgeable of PHP for server-side scripting
- Experience with Filmora and Adobe Premiere Pro for video editing

SEMINAR ATTENDED

- Google DevFest 2024 (Google Developer Group CDO)
– Nov 16, 2024